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AI Financial Advisor: Agentic Portfolio Construction

Building intelligent, personalized portfolios — and learning to question them

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DURATION
12 Weeks · 3 Phases

THE PROBLEM

Why Financial Advisory Needs Disruption

- **Human capital is your biggest asset** — yet most advisors ignore it entirely
- **Commission-driven advice** creates systematic bias toward high-margin products
- **One-size portfolios** ignore the correlation between career risk and market risk
- **Generic 60/40** is not personalized — it's an industry default

The gap: A tech executive and a biology professor have radically different risk profiles — but may receive identical allocations from a robo-advisor.

Biology Professor

Bond-like human capital
Stable, government-backed income
→ Can absorb equity risk

Tech Executive

Equity-like human capital
Correlated with tech sector
→ Must hedge tech exposure

Financial Professional

Pro-cyclical income
Falls with market downturns
→ Needs uncorrelated assets

THE THESIS

Multi-Agent AI Can Construct Better Portfolios

- A pipeline of specialized agents each handles **one well-defined problem**
- Human capital valuation is a **quantitative input**, not a qualitative guess
- Regime detection **dynamically** changes allocation — not annually
- The system produces **auditable reasoning**, not a black-box number

The distinguishing feature is the validation pipeline.
Anyone can build a robo-advisor. The hard question is: *how do you know when to trust it?*

What makes this different

Human capital as a bond-equivalent asset with present value computed from career data — not a checkbox field in a risk questionnaire

The validation pipeline

Risk and Compliance agents act as adversarial reviewers — if they approve everything, something is wrong

The research question

When does personalization actually beat a simple 80/20 portfolio? Prove it with backtesting.

SYSTEM ARCHITECTURE

The 5-Agent Pipeline

AGENT 1 — Profile

INPUT
- Client questionnaire
- Career / income data
- Household balance sheet

OUTPUT
- HC present value
- Income volatility σ
- Effective risk budget

→

AGENT 2 — Research

INPUT
- FRED macro series
- CRSP price / factor data
- Current profile

OUTPUT
- Regime classification
- Asset universe scores
- FF factor exposures

→

AGENT 3 — Allocation

INPUT
- Profile + risk budget
- Regime + asset scores
- Correlation matrix

OUTPUT
- Portfolio weights
- LLM-generated rationale
- Projected metrics

→

AGENT 4 — Risk

INPUT
- Proposed weights
- Stress scenarios
- Profile constraints

OUTPUT
- Approve / reject / revise
- CVaR, max drawdown
- Scenario P&L

→

AGENT 5 — Compliance

INPUT
- Final weights
- Regulatory rules
- Fiduciary checklist

OUTPUT
- Compliant / flag / reject
- Audit trail
- Client disclosure draft

THREE CLIENT PERSONAS

The Counterintuitive Insight: Same Wealth, Different Portfolios

Biology Professor

Tenure-track

Bond-like HC

Human capital: Stable, inflation-linked, government-backed income. Low σ, long duration.

Implication: HC already acts like bonds → portfolio CAN hold more equity risk than conventional wisdom suggests

Allocation signal: Overweight equities vs. 60/40 baseline

Tech Executive

FAANG / startup

Equity-like HC

Human capital: Comp = salary + RSUs + bonus. Highly correlated with tech sector. High β, option-like profile.

Implication: Already long tech — adding QQQ doubles the risk. Must actively hedge tech concentration.

Allocation signal: Underweight tech, overweight defensive / intl

Financial Professional

Investment banking / PE

Human capital: Pro-cyclical income — bonuses collapse in downturns. High correlation with broad market.

Implication: When markets fall, income and portfolio fall together. Needs assets that are uncorrelated or negatively correlated.

Allocation signal: Gold, TIPS, long Treasuries, alts

DATA SOURCES

What We're Working With

FRED Macro Series

`T10Y2Y` `BAMLH0A0HYM2` `UNRATE` `CPIAUCSL` `FEDFUNDS` `VIXCLS`

Yield curve, credit spreads, unemployment, inflation, fed funds rate, volatility index. Used for **regime classification**.

Fama-French Factors

`Mkt-RF` `SMB` `HML` `RMW` `CMA`

Five-factor model for explaining portfolio returns and decomposing alpha. Used in **Research Agent** for factor exposure analysis.

WRDS / CRSP

`crsp.msf_v2` `comp.funda`

Monthly security file for price/return history. Compustat for fundamentals. Used for **asset universe scoring** and backtesting.

> **Data pipeline priority (Week 3):** FRED is free and starts immediately. WRDS requires institutional access — confirm credentials in Week 1. The reference implementation uses **yfinance** as a free fallback; swap in CRSP when access is confirmed.

> **No synthetic data.** Every backtest must use actual historical prices. Regime labels must be assigned from real FRED data, not manually set.

HISTORICAL REGIMES

Five Periods That Test Every Allocation

Regime	Dates	Defining Features	Differential Persona Impact
Dot-com Bust	2000-03 – 2002-10	Tech collapse, mild recession, flight to bonds	Tech exec: HC + portfolio both crushed. Professor: protected. Fin. pro: bonus cliff.
GFC	2007-10 – 2009-03	Credit crisis, unemployment surge, systemic risk	Fin. pro: income and portfolio collapse simultaneously. Professor: minimal HC impact.
COVID Crash	2020-02 – 2020-04	Fastest bear market, V-shaped recovery, ZIRP extension	Short sharp shock — tests how quickly the system reacts. Regime detection speed matters.
Rate Hike Cycle	2022-01 – 2023-07	8% CPI, 500bp hike, bond losses, tech selloff	Professor: bond-like HC loses PV. Tech exec: RSU value drops 50%+. Critical test case.
AI Boom	2023-10 – present	AI-driven tech rally, concentration risk, NVDA >100%	Tech exec: already concentrated — system should flag overweight. Professor: opportunity.

Key test: Does your regime classifier correctly identify each period from FRED data alone? Validate labels before using them in backtesting. Garbage labels → garbage backtest.

WEEK 1

Kickoff & Assessment

Establish team roles, confirm access, and study the reference implementation — a working pipeline built by agents in a few hours.

- Introductions — backgrounds, strengths, preferred roles
- Walk through the full project scope: 5 agents, 3 personas, 12 weeks
- Assign agent ownership: Aidan → Allocation + Risk, James → Profile + Research (lead), Neha → Compliance + PM
- Weekly meetings: **Fridays, 10:00 AM ET**
- Review grading rubric and reference implementation

A complete 5-agent pipeline was built by AI agents in one afternoon. It is the floor — not the ceiling. Your job is to go beyond it.

HOMEWORK — DUE WEEK 2

- → Study the vibed implementation — clone it, run it, trace one persona end-to-end
- → Each student: write up 3 things the vibed version does wrong for your agent area
- → Confirm WRDS credentials and run one test query on crsp.msf_v2
- → Set up shared GitHub repo with agreed branch strategy

WEEK 2

Architecture Review

Converge on a shared data schema and define explicit contracts between every agent pair. The vibed implementation has one — adopt or revise it.

- Each student presents: what the vibed version gets wrong in their agent area
- Define inter-agent contracts: data types and formats between every agent pair
- Agree on the human capital PV model: variables, formula, discount rate
- Decide LLM choice, API structure, how agents call each other
- Set up dev environment: Python, dependencies, testing framework

The contract matters more than the code. If agents disagree on what a "portfolio weight" is (percent? decimal? basis points?), the system silently produces wrong results.

HOMEWORK — DUE WEEK 3

→ Finalize and commit the inter-agent contract as a typed Python dataclass or Pydantic schema

→ Document the human capital PV formula with all variable definitions

→ Write one integration test that validates data flows from Profile → Research

→ Read: Merton (1971) on intertemporal portfolio theory

WEEK 3

Data Pipeline & Regimes

Get clean, reproducible data flowing from FRED and WRDS, and define each historical regime with explicit date ranges.

- Build FRED data pipeline: pull T10Y2Y, BAMLH0A0HYM2, UNRATE, CPIAUCSL, FEDFUNDS, VIXCLS
- Build WRDS pipeline: crsp.msf_v2 monthly returns + comp.funda fundamentals
- Assign regime labels to historical periods with exact date boundaries
- Validate: does the data "look right" for each regime? GFC spreads spike, VIX spikes in COVID
- Discuss regime classification strategies: rule-based vs. ML-based vs. LLM-classified

Watch out: FRED series have different frequencies and update lags. Align everything to monthly before joining. Missing data → forward-fill or flag, never silently drop.

HOMEWORK — DUE WEEK 4

→ Deliver a clean Parquet file with all FRED series, monthly, no gaps

→ Deliver regime label CSV: start_date, end_date, label for all 5 regimes

→ Spot-check: pull VIXCLS for March 2020 — confirm spike > 80

WEEK 4

Profile Agent & Human Capital

Profile Agent producing meaningfully different outputs for each of the three personas — not just different numbers, different decisions.

- Implement the human capital PV model: discount future income by risk-adjusted rate
- Profile Agent demo: same age/wealth, three personas → three different risk budgets
- Discuss: what variables actually change the allocation? Sensitivity analysis.
- Review: does the professor output look "bond-like"? Does the tech exec flag tech correlation?

The sanity check: Run all three personas through Profile Agent. If the outputs are similar, the human capital model is broken or underweighted.

HOMEWORK — DUE WEEK 5

→ Profile Agent: working, tested, returns typed output for all 3 personas

→ HC sensitivity table: vary income growth ±2% — show effect on PV and risk budget

→ Unit tests for Profile Agent with at least 5 persona variations

1 PHASE 1 GATE | Profile Agent + Data Pipeline reviewed and signed off

WEEK 5

Research Agent & Regime Classification

> *Validate that the Research Agent's regime classifier correctly identifies known historical periods from macro data alone.*

- Implement Research Agent: ingest FRED series → classify current regime → score asset universe
- Backtest the classifier: feed it 2000–2024 data, compare output labels to ground truth
- Compute Fama-French factor exposures for each asset in the universe
- Test: ask Research Agent "what regime is 2008-10?" — it should say GFC
- Discuss: how sensitive is classification to feature selection?

Regime detection is load-bearing. If the classifier is wrong, every downstream allocation is wrong. Validate against known regimes before trusting it on new data.

HOMEWORK — DUE WEEK 6

→ Regime classifier confusion matrix across all 5 regimes
→ Research Agent: returns factor exposures + regime label for a given date
→ Asset universe: minimum 20 tickers with monthly return history from CRSP

WEEK 6

Allocation Agent & First Pipeline Run

Run the system end-to-end for one persona — inputs flow from Profile through Research into Allocation and produce real portfolio weights.

- Build Allocation Agent: combine risk budget + factor scores + regime → optimize weights
- Define the LLM's role: generating rationale text, or actually influencing weights?
- Run first end-to-end pipeline: Biology Professor persona, current regime
- Review output: does the allocation make intuitive sense? Are weights sum to 1?
- Discuss: mean-variance optimization vs. risk parity vs. LLM-directed allocation

LLM clarity question: Is the LLM telling the optimizer what to do, or explaining what the optimizer already did? Be explicit. The paper needs to answer this.

HOMEWORK — DUE WEEK 7

→ End-to-end pipeline working for Biology Professor: Profile → Research → Allocation
→ Allocation Agent: tested, weights sum to 1, respects constraints
→ Document the LLM's exact role in the allocation decision — one precise paragraph

WEEK 7

Risk & Compliance: The Adversarial Layer

Build the agents designed to say "no" — and verify that they actually do.

- Risk Agent: implement CVaR, max drawdown, and at least 3 stress scenarios per regime
- Compliance Agent: fiduciary checklist, concentration limits, disclosure requirements
- Test: deliberately pass a bad portfolio (100% equities for a retired client) — does Risk reject it?
- Test: deliberately violate a concentration rule — does Compliance flag it?
- Implement the feedback loop: Allocation Agent revises weights when Risk/Compliance rejects

Critical calibration test:
Run all three personas through the full pipeline. If Risk Agent approves every single portfolio on the first pass, your thresholds are too lenient — or the adversarial logic isn't working.

HOMEWORK — DUE WEEK 8

→ Risk Agent: rejects at least 1 out of 3 persona portfolios in initial run

→ Compliance Agent: correctly flags concentration violation in a test case

→ Document rejection criteria and revision feedback loop logic

WEEK 8

Full System Demo & Backtesting

Live demo of all five agents across all three personas — then launch the backtesting framework.

- Live demo: run all three personas through the complete 5-agent pipeline
- Each student presents their agent: what it does, how it decides, example output
- Compare the three output portfolios — are they meaningfully different?
- Design the backtesting framework: walk-forward, monthly rebalance, transaction costs
- Begin running backtests on the 5 historical regimes

Demo standard: The three persona portfolios should be visibly, meaningfully different. If Professor and Tech Exec get similar equity allocations, the human capital model is not driving decisions.

HOMEWORK — DUE WEEK 9

→ Backtesting framework: produce return series for each persona over 2000–2024
→ Benchmark comparison: run the same framework with a simple 80/20 portfolio
→ Begin writing: "System Architecture" section of the paper

2 PHASE 2 GATE | Full system demo + backtesting framework reviewed and signed off

WEEK 9

Backtesting Results

Interpret the backtests honestly — when does the system beat a simple benchmark, and when doesn't it?

- Review full backtest results: Sharpe ratio, max drawdown, CAGR per persona per regime
- The central question: **when does personalization beat 80/20?**
- Evaluate regime detection timing: how many months late is the classifier?
- Attribution analysis: is outperformance from human capital, regime, or factor tilts?
- Identify failure modes: periods where the system underperforms — and why

Honest results matter more than good results. If the system doesn't beat 80/20 in most regimes, say so — and explain why. That is a publishable finding. Cherry-picking is not.

HOMEWORK — DUE WEEK 10

→ Backtest result table: Sharpe, CAGR, max drawdown for each persona × regime

→ Identify the one regime where your system adds the most value — explain why

→ Identify the one regime where it underperforms — explain why

→ Draft "Results" section of the paper

WEEK 10

The Trust Question

No new code this week. Step back and critically evaluate the system you built. This is where the research becomes real.

- **Hallucination detection:** When does the LLM generate confident but wrong rationale?
- **Fiduciary duty:** Would a licensed advisor be comfortable signing off on this output?
- **Liability:** If the portfolio loses 40%, who is responsible? The system? The student? The advisor?
- **Overfitting risk:** Was the regime classifier tuned on data that includes the test period?
- **Robustness:** Change one parameter — does the portfolio change

This is not optional introspection. Every committee member at your defense will ask a version of: "Why should I trust this?" If you don't have a prepared answer, the defense will fail regardless of code quality.

HOMEWORK — DUE WEEK 11

→ Write a 1-page "trust evaluation": 3 specific ways the system can be wrong

→ Identify one instance of LLM-generated rationale that is misleading or overconfident

→ Draft "Limitations and Future Work" section of the paper

WEEK 11

Paper Draft Review

The paper must be defensible: every claim backed by data, every design choice justified, every limitation acknowledged.

- Full draft circulated 48 hours before the meeting — come with written comments
- Structure review: Introduction → Related Work → Methodology → Results → Discussion → Limitations
- Substance check: are backtest tables complete? Do results match what the paper claims?
- Practice answering: "What would break this system in live deployment?"
- Practice answering: "Why not just use a target-date fund?"

Defense questions to prepare for:
(1) What is the incremental contribution over existing robo-advisors?
(2) How did you validate the human capital model?
(3) What happens when the regime classifier makes a mistake?

HOMEWORK — DUE WEEK 12

→ Final paper draft (12–15 pages): all sections complete, figures included
→ Each student: prepare a 5-minute presentation segment on their agent
→ Prepare answers to the three defense questions above — in writing

WEEK 12

Defense Prep: Mock Defense

Simulate the formal defense. Each student presents. Hard questions follow. Iterate until it's ready.

- Each student: 5-minute presentation on their agent — no slides from Week 1, fresh material
- Full Q&A: 15 minutes of adversarial questions from the mentor
- Review: is the system coherent as a whole, not just three separate agents?
- Dry-run the demo: does the live pipeline still run? Does it produce the same results as the paper?
- Final paper: submit polished, proofread, formatted version

Success criteria: After mock defense, each student can answer any question about any part of the system — not just their own agent. Integration understanding is what separates a good capstone from a great one.

FINAL DELIVERABLES — ALL DUE AT DEFENSE

→ Complete codebase: clean, documented, reproducible
→ Final paper: 12–15 pages, all results included
→ Presentation slides for formal defense
→ Working demo: can be run live in 5 minutes on a fresh machine

3 **PHASE 3 GATE | Mock defense passed — cleared for formal submission**

DELIVERABLES

Everything Due at Final Defense

SYSTEM

- [ ] 5-agent system: Profile, Research, Allocation, Risk, Compliance
- [ ] 3 persona portfolios: Professor, Tech Exec, Financial Pro
- [ ] Working end-to-end pipeline (reproducible, no manual steps)
- [ ] Inter-agent contracts: typed, documented, tested
- [ ] Regime classifier: validated against 5 historical periods

ANALYSIS

- [ ] Regime backtesting: all 5 periods × 3 personas
- [ ] Benchmark comparison: AI system vs. 80/20 portfolio
- [ ] Agent vs. industry robo-advisor comparison

RESEARCH

- [ ] Trust evaluation: 3 specific failure modes documented
- [ ] Hallucination detection: at least 1 identified example
- [ ] 12–15 page research paper (submission-quality)

DEFENSE

- [ ] Oral defense: each student presents 5 minutes
- [ ] Live demo during defense (system runs in real time)
- [ ] Q&A: each student can speak to the full system

> **Phase gates:** W4 (Phase 1) · W8 (Phase 2) · W12 (Phase 3)
> Work not meeting a gate is addressed before proceeding.

"The goal isn't to build a robo-advisor that works. The goal is to build one — then figure out **why you shouldn't trust it.**"

That second part is where the research lives.
Anyone can build the system. Not everyone can evaluate it honestly.

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